S&S QUIZ-1 SOLUTION

Sol 1(a):- For Periodic signal,

$$x(t+T) = x(t) \quad \forall t$$

(0.5 Mark)

where, T = FTP (smallest positive)

Given, $x(t) = e^{j(\pi t - 1)}$ — (1)

$$x(t+T) = e^{j[\pi(t+T)-1]}$$

$$= e^{j(\pi t - 1)} \cdot e^{j\pi T} \quad \text{— (2)}$$

In eqn (2), if $T = 2$ then — $e^{j\pi T} = e^{j2\pi} = 1$

Hence, $x(t+T) = x(t)$

$\therefore$ $x(t)$ is periodic signal with FTP, $T = 2$ → (2 Mark)

Sol 1(b):- For Periodic signal,

$$x[n+N] = x[n] \quad \forall n$$

(0.5 Mark)

where, N = FTP ∈ I & smallest value

Given, $x[n] = \cos\left(\frac{\pi}{8}n^2\right)$ — (1)

$$x[n+N] = \cos\left[\frac{\pi}{8}(n+N)^2\right]$$

$$= \cos\left[\frac{\pi}{8}n^2 + \frac{\pi}{8}N^2 + \frac{2\pi}{8}nN\right] \quad \text{— (2)}$$

As we know that — $\cos(2\pi \pm \theta) = \cos\theta$

In eqn (2), $\frac{\pi}{8}N^2 + \frac{2\pi}{8}nN = 2\pi k$

smallest integer value for which above condition satisfy is — $N = 8$

Hence, $x[n+N] = x[n]$

$\therefore$ $x[n]$ is periodic signal with FTP, $N = 8$ → (2 Mark)

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Sol 2(a):- Given that –

$$x[n] = -1 + \sum_{K=2}^{\infty} \delta[n-K]$$

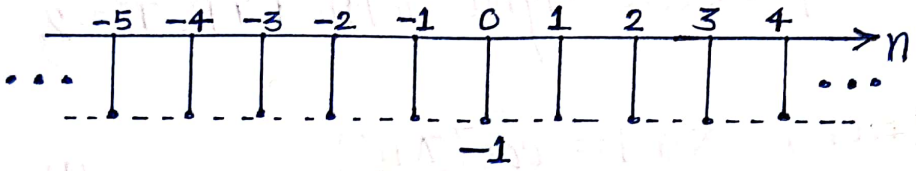
$$= x_1[n] + x_2[n]$$

where,

$$x_1[n] = -1$$

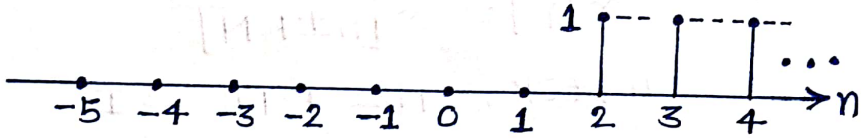
$$x_2[n] = \sum_{K=2}^{\infty} \delta[n-K]$$

$x_1[n]$



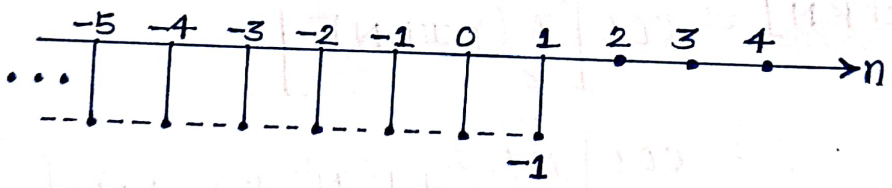
→ (0.5 M)

$x_2[n]$



→ (0.5 M)

$x[n]$



→ (1.5 M)

Sol 2(b):- By the above plot of $x[n]$, we can get –

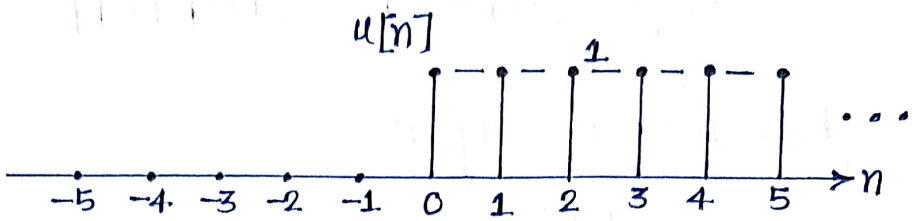
$$x[n] = \begin{cases} -1, & n \leq 1 \\ 0, & n \geq 2 \end{cases}$$

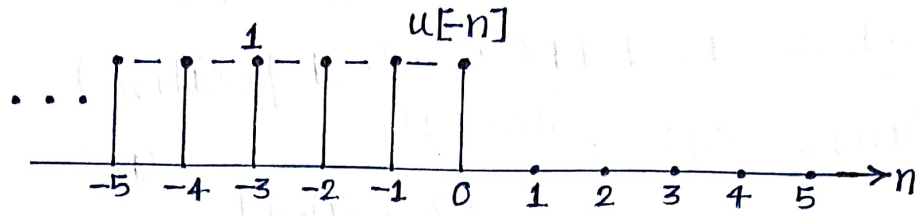
$$= -u[-(n-1)]$$

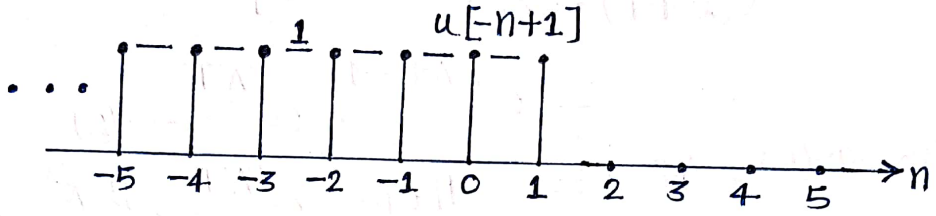
$$= -u[-n+1] = A\,u[Mn+n_0]$$

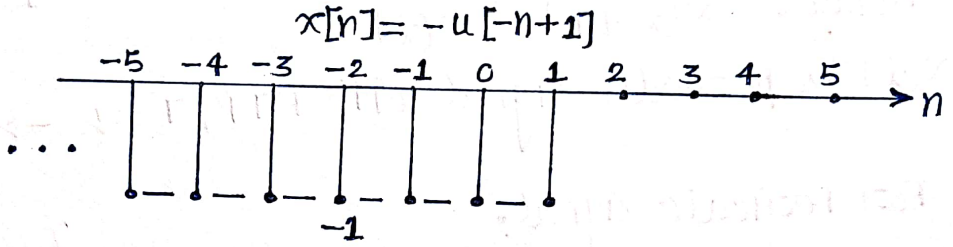
Hence, $A = -1$, $M = -1$, $n_0 = 1$ → (3 X 0.5 Mark)

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u[n]
1
-5 -4 -3 -2 -1 0 1 2 3 4 5
n


u[-n]
1
-5 -4 -3 -2 -1 0 1 2 3 4 5
n


u[-n+1]
1
-5 -4 -3 -2 -1 0 1 2 3 4 5
n


x[n] = -u[-n+1]
-5 -4 -3 -2 -1 0 1 2 3 4 5
n
-1

→ (4 × 0.25 Mark)